

Data Descriptors

CMPSC 105 – Data Exploration



ALLEGHENY COLLEGE

Goals

- Central Tendencies
 - Mode, Median, Mean
- Distributions
 - Histograms, Range, Quartiles, Standard Deviation, Z-scores
- Stats in Practice
- Pandas
- Class Activity

Measures of Central Tendency

Measures of Central Tendency

Mode Definition

The most common value in a sample. Usually, a measure of “central tendency”.

Mode Examples

3, 4, 5, 6, 7, 7, 7, 8, 8, 9

7

8, 5, 3, 7, 4, 6, 7, 8, 9, 7

7

3, 4, 5, 6, 7, 7, 7, 8, 8, 8, 9

Use the average in case of multiple modes, or a set with both modes

7.5 or {7,8}

Measures of Central Tendency

Median Definition

Middle value once a variable has been sorted low to high

Median Examples

raw: 3, 4, 7, 2, 3, 7, 4, 2, 4, 7, 4

sorted: 2, 2, 3, 3, 4, 4, 4, 4, 7, 7, 7

6th is middle of 11: 4

Take mean if there are two middles

Mean Definition

Sum of all the values in a sample divided by the number of values.

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$$

Measures of Distributions

Range Definition

Distance between lowest and highest values in a variable
(subtract high minus low)

- raw: 3, 4, 7, 2, 3, 7, 4, 2, 4, 7, 4
- sorted: 2, 2, 3, 3, 4, 4, 4, 4, 7, 7, 7
- highest: 7
- lowest: 2
- range: $7 - 2 \rightarrow 5$

Quartiles Definition

The middle value (median), the middle value of the lower half (lower quartile), the middle value of the upper half (upper quartile)

- raw: 3, 4, 7, 2, 3, 7, 4, 2, 4, 7, 4
- sorted: 2, 2, 3, 3, 4, 4, 4, 4, 7, 7, 7
- middle is element 6 out of 11: 4
- lower quartile is element 3 out of 5: 3
- upper quartile is element 3 out of 5: 7
- exclude the median in calculating the uq and lq

Measures of Distributions

Box Plots

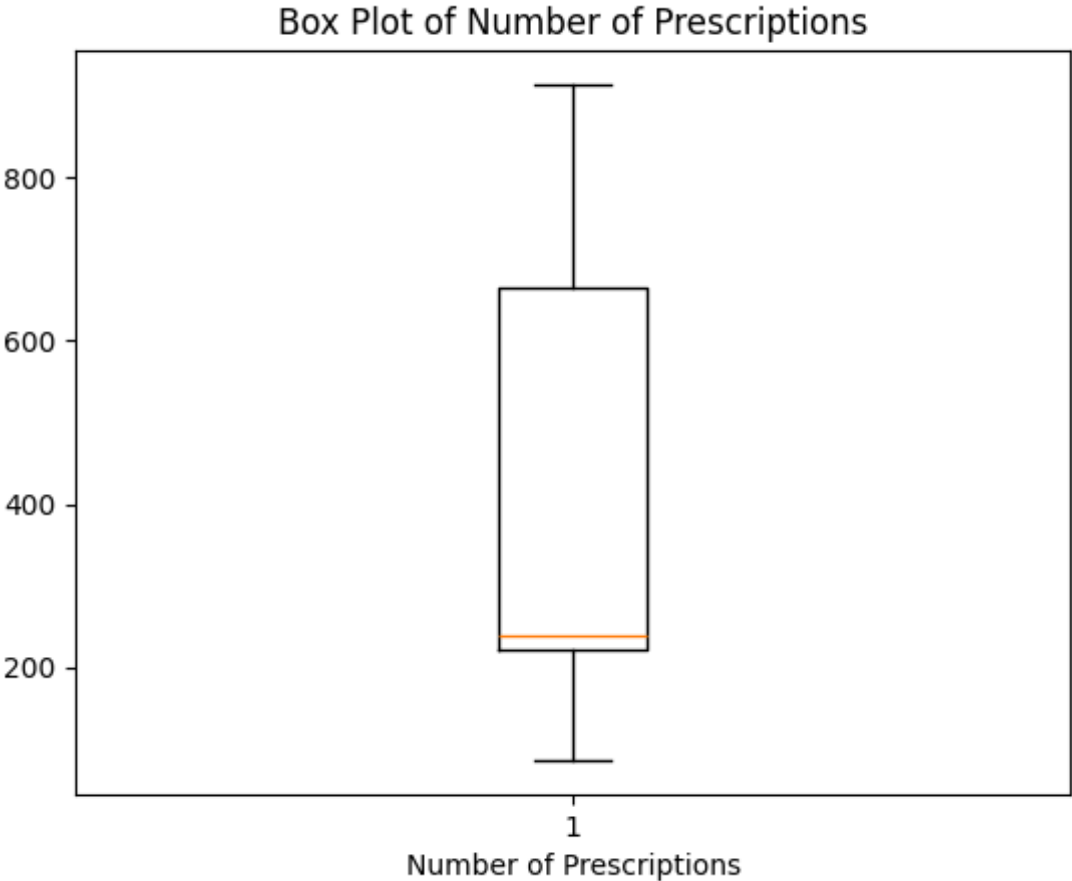
Visualizations of quartiles and range (and mean and outliers)

Box Plots

Sketch a box plot for the following data:

- raw: 3, 4, 7, 2, 3, 7, 4, 2, 4, 7, 4
- sorted: 2, 2, 3, 3, 4, 4, 4, 4, 7, 7, 7
- middle is element 6 out of 11: 4
- lq is element 3 out of 5: 3
- uq is element 3 out of 5: 7

Box and Whisker Plot



Variance Definition

Describes the spread of the data and how tightly clustered around the mean it is.

- Mathematically, it is the normalized sum of squared errors.
- Errors are comparisons of each data point with the mean
- (subtraction)
- Squaring makes all errors positive and penalizes large errors
- Summing and normalizing is like finding the average squared error
- σ^2 represents variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}$$

Standard Deviation Definition

Square root of variance. Describes the spread of the data and how tightly clustered around the mean it is.

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

Visualizations of Standard Deviation

- small STD $\rightarrow$ narrow distribution of data, most is close to the mean
- large STD $\rightarrow$ wide distribution of data, a lot is not close to the mean
- x axis is the variable (e.g. birth weight), y axis is count or frequency or proportion (how many babies born with that weight)

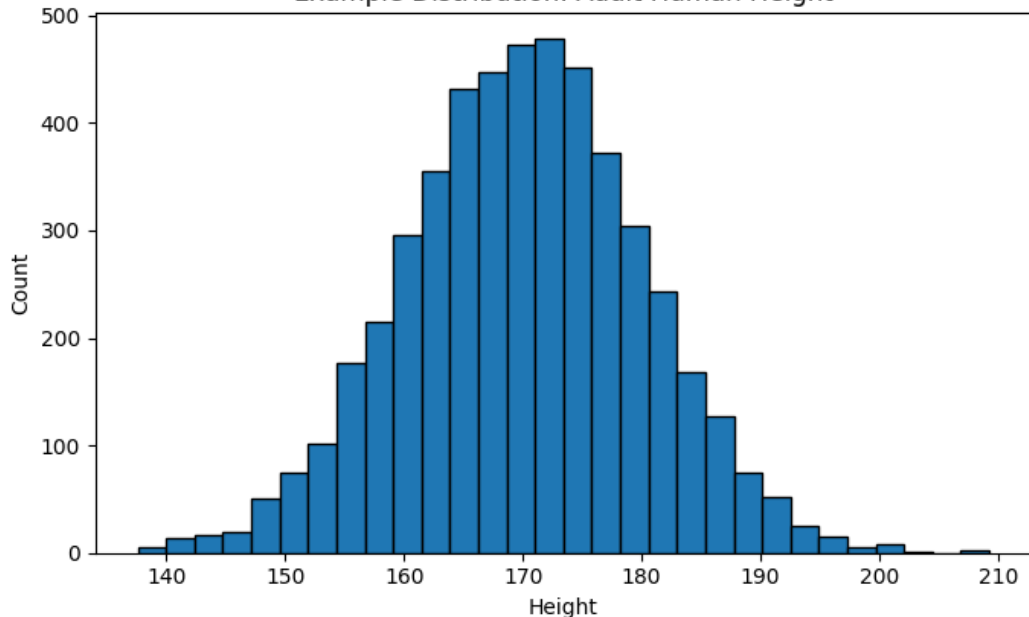
Visualization of Distributions

Histograms show spread as well as distribution shape.

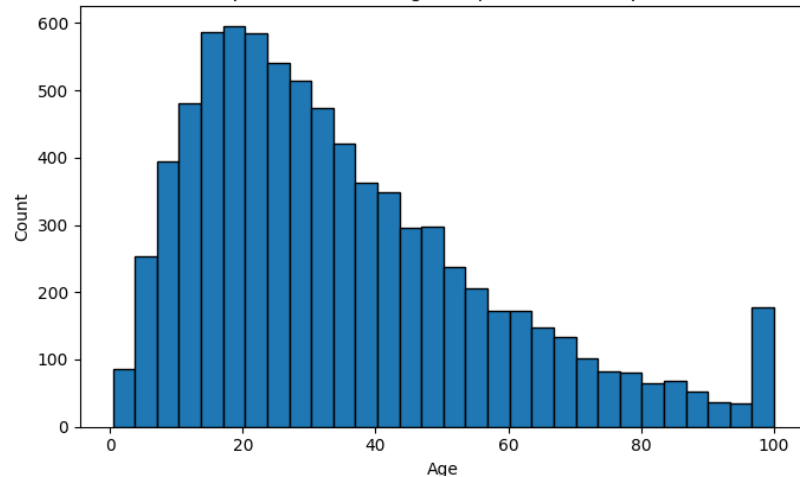
- Which distribution might describe adult human height?
- Which distribution might describe age frequencies in the human population?
- Which distribution might describe median household income in USA?

Visualization of Distributions

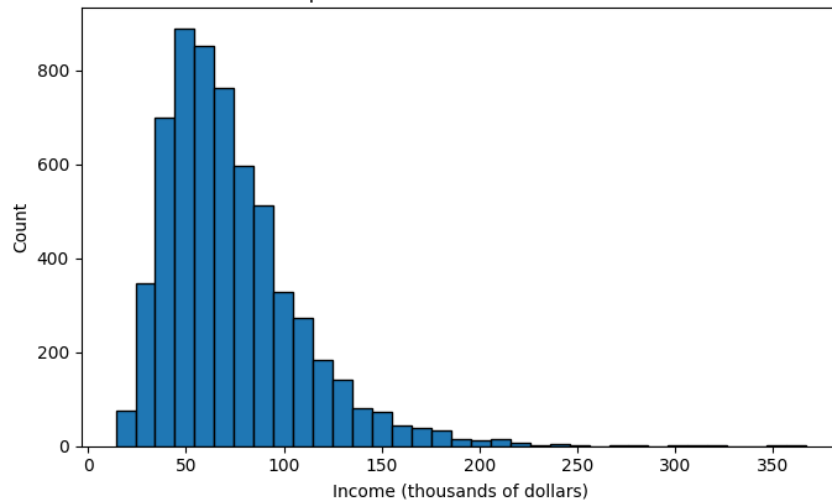
Example Distribution: Adult Human Height



Example Distribution: Age Frequencies in a Population



Example Distribution: Household Income



Note: The distributions in this figure were generated for illustration purposes by the instructor and are not derived from real datasets.

Normal Distribution Terminology Overload

- Normal
- Bell-Shaped
- Gaussian
- Standard Normal (mean 0, std 1)

Z-scores

A linear transformation that converts data points into measures of how many standard deviations away the data is from the mean.

- data point is compared with mean
- the difference is normalized with respect to the standard deviation (division)
- z-score of 0 means that the data point is identical to the mean
- negative z-score means that the data point is smaller than the mean
- positive z-score means that the data point is larger than the mean

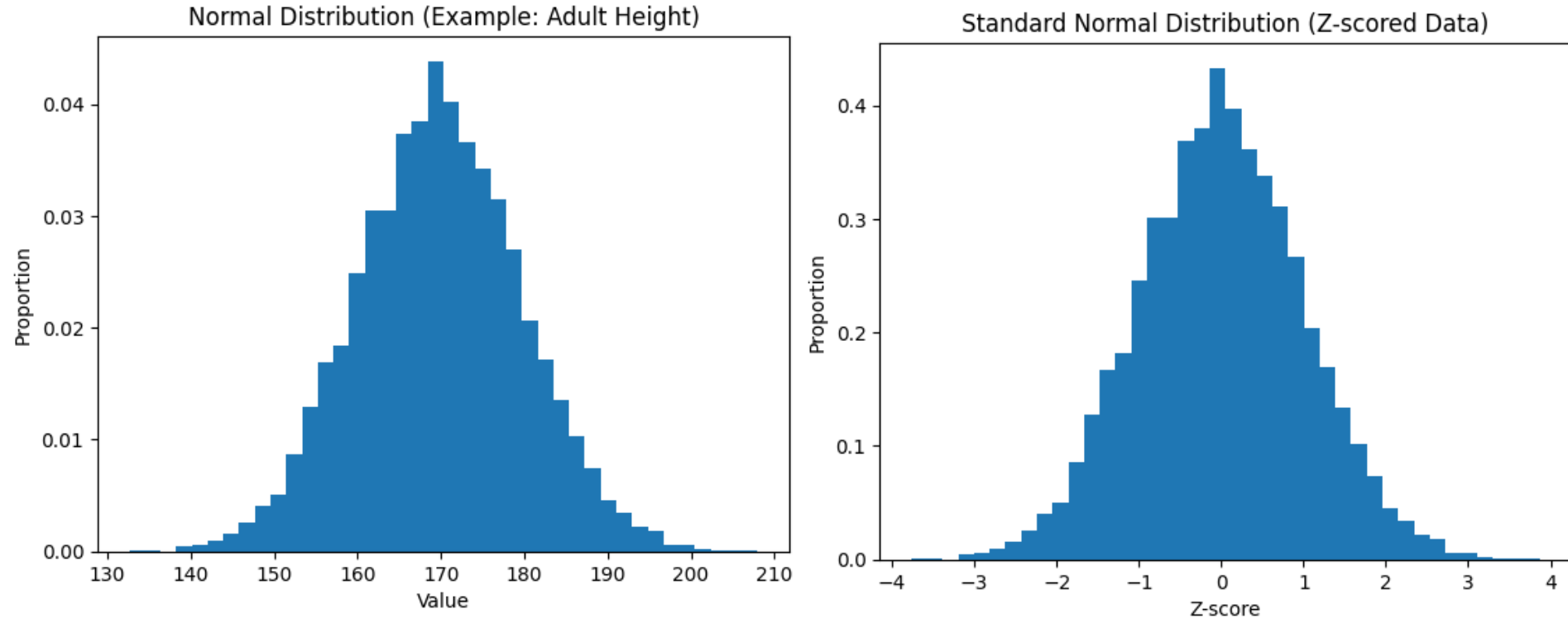
$$Z_i = \frac{x_i - \bar{x}}{\sigma}$$

Z-scored Normal Data

When normal (gaussian) data is z-scored, the distribution looks like this:

- x axis is the z-score of normal data
- y axis is the proportion of data with that z-score
- area totals 1, or 100%
- the standard deviation of z-scored data is always 1
- the mean of z-scored data is always 0

Z-scored Normal Data



Note: *The distributions in this figure were generated for illustration purposes by the instructor and are not derived from real datasets.*

Stats in Practice

Pandas Examples

- mean - `df.mean()` or `df['column'].mean()`
- median - `df.median()` or `df['column'].median()`
- mode - `df.mode()` or `df['column'].mode()`
- std - `df.std()` or `df['column'].std()`
- z-score - `(df['column'] - df['column'].mean()) / df['column'].std()`
- histograms

```
import matplotlib.pyplot as plt  
plt.hist(variable, bins)
```